

Course Notes: Pearson Edexcel International A Level Physics

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Chapter 1

Physics Year 1

Consider a constant force $\vec{F}$ causing a ball of mass m which is moving at a velocity $\vec{v}$ to come to rest over a distance $\vec{s}$.

TODO: TikZ diagram

Using SUVAT:

$$v^2 = u^2 + 2as \quad (1.1)$$

But since it comes to rest, $v = 0$, and the initial velocity $u = |\vec{v}|$. Rearranging for a :

$$\begin{aligned} 0 &= v^2 + 2as \\ v^2 &= -2as \\ -\frac{v^2}{2} &= as \\ as &= -\frac{v^2}{2} \end{aligned} \quad (1.2)$$

Since the work done W by a constant force is given by:

$$W = \vec{F} \cdot \vec{s} = m\vec{a} \cdot \vec{s} \quad (1.3)$$

We can substitute equation 1.2 into this to get something already resembling the kinetic energy formula:

$$\begin{aligned} W &= m \cdot -\frac{v^2}{2} \\ W &= -\frac{1}{2}mv^2 \end{aligned}$$

Since the work done W is equal to the change in kinetic energy, and the particle ends at rest, we have:

$$W = E_{kf} - E_{ki}$$

$$W = 0 - E_{ki}$$

$$W = -E_{ki}$$

$$E_{ki} = -W$$

$$E_{ki} = \frac{1}{2}mv^2$$

Chapter 2

Physics Year 2

2.5 Further Mechanics

2.5B Circular Motion

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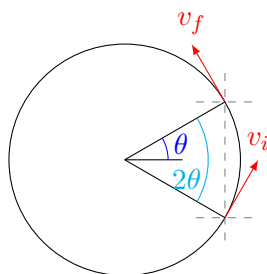
Proof of acceleration formula during Uniform Circular Motion

During uniform circular motion, an object moves in a circle of radius r at a constant speed v and constant angular velocity ω . Although the speed is constant, the direction of the velocity vector is continuously changing, which means the object is accelerating.

The equation for the magnitude of the centripetal acceleration a is given by:

$$a = \frac{v^2}{r} \quad (2.1)$$

Here is a proof of this formula, with help from a diagram.



By definition,

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} \quad (2.2)$$

Noticing that since there is no change in the y-component of v_i and v_f , the acceleration in the y-direction is zero. Therefore, we only need to consider the x-component of the velocity. Thus:

$$a = a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{v_{fx} - v_{ix}}{\Delta t}$$

By trigonometry, we can see that:

$$v_{ix} = v \sin \theta$$

$$v_{fx} = -v \sin \theta$$

And from the definition of angular velocity, we know that:

$$\omega = \frac{\Delta \theta}{\Delta t}$$

$$\omega = \frac{2\theta}{\Delta t}$$

$$\Delta t = \frac{2\theta}{\omega}$$

Now, work out the limit:

$$a = \lim_{\Delta t \rightarrow 0} \frac{-v \sin \theta - v \sin \theta}{\Delta t}$$

$$a = \lim_{\Delta t \rightarrow 0} -\frac{2v \sin \theta}{\Delta t}$$

$$a = -\lim_{\theta \rightarrow 0} \frac{2v\omega \sin \theta}{2\theta}$$

$$a = -\lim_{\theta \rightarrow 0} v\omega \cdot \frac{\sin \theta}{\theta}$$

$$a = -v\omega \cdot \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$$

$$a = -v\omega \cdot 1$$

$$a = -v\omega \cdot 1$$

Now finally, since $v = r\omega$, we can substitute for ω :

$$a = -v \cdot \frac{v}{r}$$

$$a = -\frac{v^2}{r}$$

Now for completeness, this is actually the acceleration in the x-direction. In this case, the x-direction is towards the center of the circle, so the negative sign indicates that the acceleration is centripetal. But the magnitude of the acceleration is of course positive.

2.6 Electric and Magnetic Fields

2.6C Electromagnetic Effects

Gravitational, electric and magnetic fields are all vector fields. The strength of each field is defined as the force per unit something experienced by an object placed in the field:

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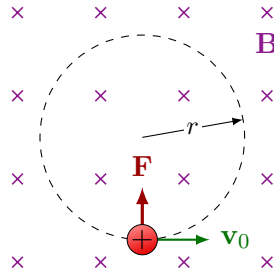
$$\vec{g} = \frac{\vec{F}}{m} \quad \vec{E} = \frac{\vec{F}}{q} \quad \vec{B} = \frac{\vec{F}}{q\vec{v}}$$

$$\text{Or } V_{\text{grav}} \nearrow \Phi = \frac{W}{m} \quad V = \frac{W}{q} \quad ? = \frac{W}{q\vec{v}}$$

$$W = \vec{F} \cdot \vec{s}$$

$$W = |\vec{F}| |\vec{s}| \cos \theta$$

Magnetic fields can cause circular motion!



The forces involved are the magnetic force and the centripetal force:

$$\vec{F}_B = q\vec{v} \cdot \vec{B} \quad \vec{F}_{cr} = \frac{m\vec{v}^2}{r}$$

Equating the magnetic force to the centripetal force, we can solve for the radius of the circular path:

$$\begin{aligned} q\vec{v} \cdot \vec{B} &= \frac{m\vec{v}^2}{r} \\ q\vec{B} &= \frac{m\vec{v}}{r} \\ r &= \frac{m\vec{v}}{q\vec{B}} \\ r &= \frac{\vec{p}}{q\vec{B}} \end{aligned}$$

See also

- Raycast AI Chat, Gemini 2.5 Pro: Electromagnetic Induction and Magnetic Forces: Equations, hints, and problem-solving

Magnetic flux through an area A in a magnetic field B is given by:

$$\Phi = \vec{B} \cdot \vec{A}$$

$$\Phi = |\vec{B}| |\vec{A}| \cos \theta$$

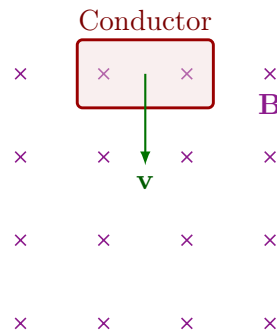
Where Φ is in webers, Wb, or T m^2 , since B is in teslas and A is in square metres.

More magnetic flux stuff:

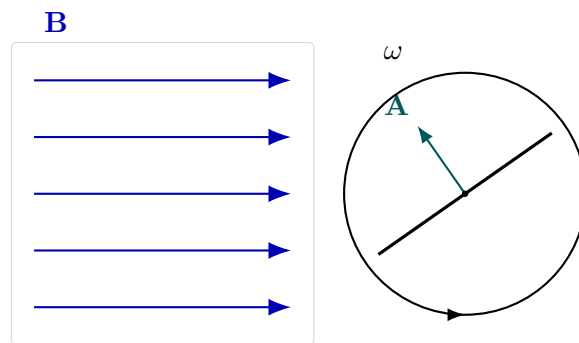
$$\frac{d}{dt} \Phi = V$$

$$\frac{d}{dt} N\Phi = V$$

“The rate of change of flux linkage through a coil gives the induced e.m.f. across the coil.”



Or we can have induction with a rotating coil in a magnetic field:



Here, e.m.f. is given by:

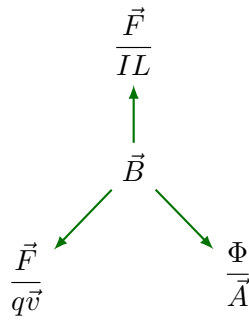
$$\mathcal{E} = \frac{d}{dt} [N\Phi] \quad (2.3)$$

Taking out the constants, rearranging and substituting:

$$\begin{aligned}\mathcal{E} &= N \frac{d\Phi}{dt} \\ \mathcal{E} &= N \frac{d}{dt} (\vec{B} \cdot \vec{A}) \\ \mathcal{E} &= N \frac{d}{dt} (|\vec{B}| |\vec{A}| \cos \theta)\end{aligned}$$

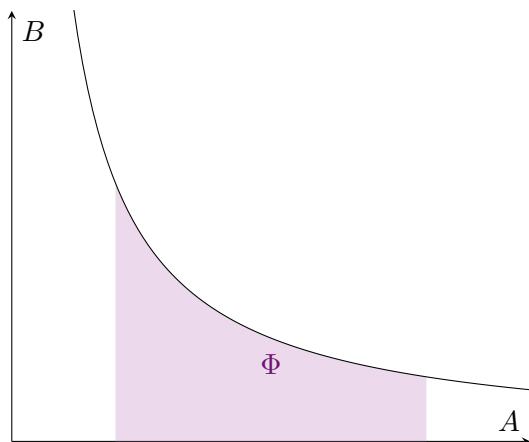
Using $A = |\vec{A}|$ and $B = |\vec{B}|$ for simplicity:

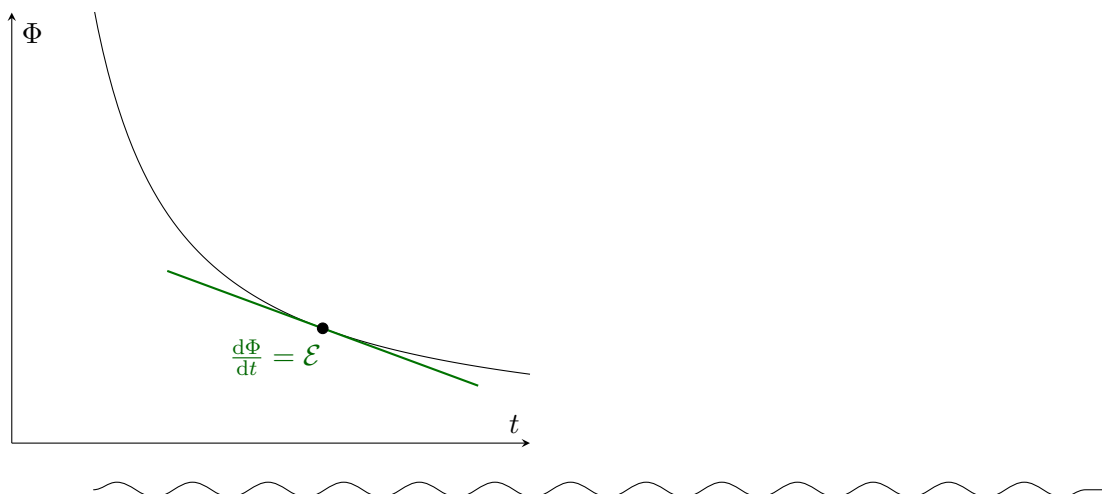
$$\begin{aligned}\mathcal{E} &= N \frac{d}{dt} (BA \cos \theta) \\ \mathcal{E} &= NBA \frac{d}{dt} [\cos \theta] \\ \mathcal{E} &= NBA \frac{d}{dt} [\cos \omega t] \\ \mathcal{E} &= -NBA\omega \sin(\omega t)\end{aligned}$$



Therefore,

$$\begin{aligned}\vec{F} &= \vec{B}IL \\ \vec{F} &= q\vec{v} \cdot \vec{B}\end{aligned}$$



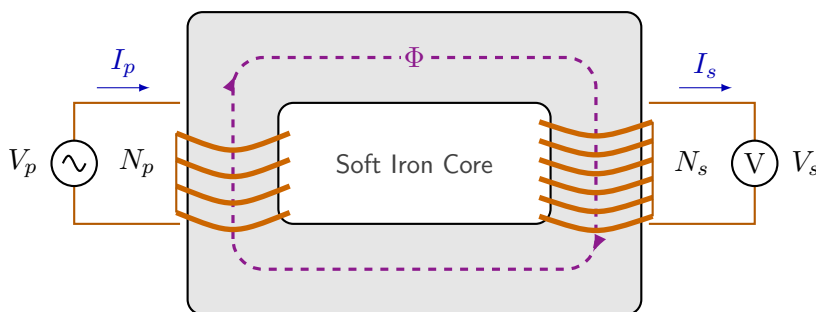


TRANSFORMERS!!!

Ratio of primary to secondary e.m.f.s is equal to the ratio of the number of turns in the primary to the secondary coil.

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \quad (2.4)$$

This equation can be derived using the following diagram of a transformer and the definition of e.m.f. seen in equation 2.3:



From the diagram, we can see that the magnetic flux Φ through each coil is the same. Therefore, we can write the e.m.f.s in each coil as:

$$\mathcal{E}_p = -N_p \frac{d\Phi}{dt}, \quad \mathcal{E}_s = -N_s \frac{d\Phi}{dt} \quad (2.5)$$

Taking the ratio of these two equations gives us equation 2.4:

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \quad (2.4 \text{ revisited})$$

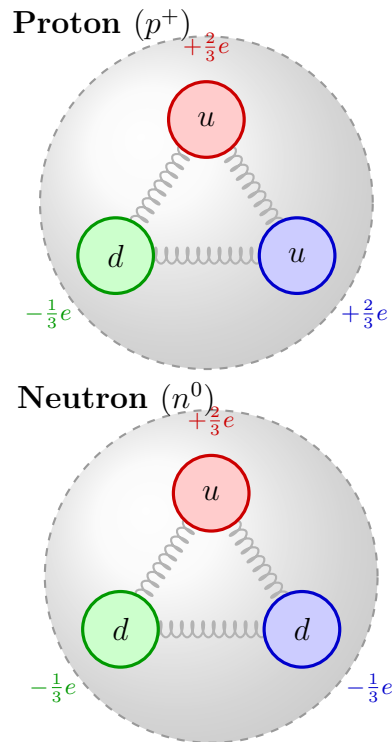
2.7 Nuclear and Particle Physics

2.7C The Particle Zoo

“Hadrons are subatomic particles made of quarks, which experience a Strong Nuclear Force.”

“Quark flavour changing is done by the Weak Nuclear Force.”

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2.8 Thermodynamics

2.8A Heat and Temperature

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Ideal Gas Behaviour

Boyle's Law In a gas, when n and T are kept constant, the pressure p is inversely proportional to the volume V :

$$p \propto \frac{1}{V} \quad (2.6)$$

2.10 Oscillations

2.10A Oscillations

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$$F_R \propto -x$$

$$a_R \propto -x$$

$$\frac{dv}{dt} \propto -x$$

$$\frac{d^2x}{dt^2} \propto -x$$

$$\frac{d^2x}{dt^2} = -cx$$

$$\frac{d^2x}{dt^2} + cx = 0$$

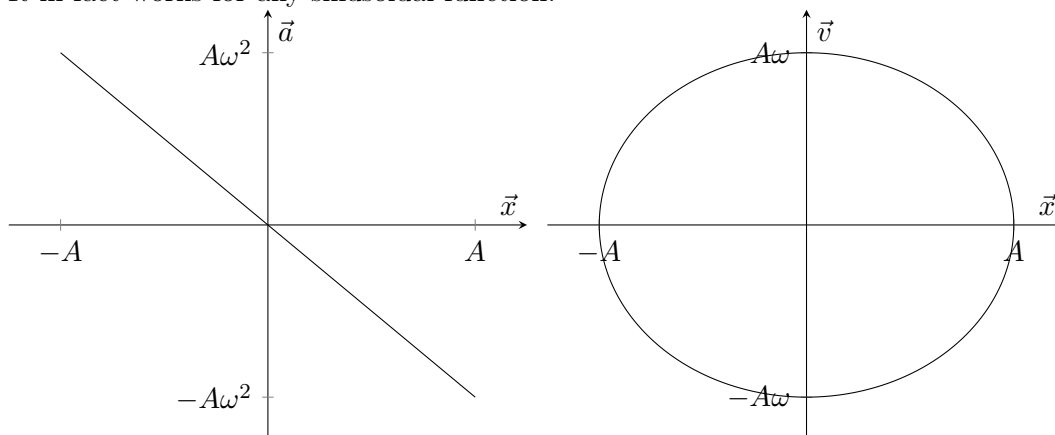


$$\begin{aligned}\vec{a}_R &= -\omega^2 \vec{x} \\ \frac{d^2 \vec{x}}{dt^2} &= -\omega^2 \vec{x} \\ \frac{d^2 x}{dt^2} + \omega^2 x &= 0\end{aligned}$$

Try $\vec{x} = A \sin(\omega t)$:

$$\begin{aligned}\frac{d^2}{dt^2}[A \sin(\omega t)] + \omega^2(A \sin(\omega t)) &= 0 \\ \frac{d}{dt}[A\omega \cos(\omega t)] + \omega^2 A \sin(\omega t) &= 0 \\ -A\omega^2 \sin(\omega t) + A\omega^2 \sin(\omega t) &= 0 \\ 0 &= 0\end{aligned}$$

It in fact works for any sinusoidal function.



Page 154 checkpoint 3:

$$\begin{aligned}
T &= 2\pi\sqrt{\frac{l}{g}} \\
\frac{T}{2\pi} &= \sqrt{\frac{l}{g}} \\
\left(\frac{T}{2\pi}\right)^2 &= \frac{l}{g} \\
l &= g\left(\frac{T}{2\pi}\right)^2 \\
l &= \frac{gT^2}{4\pi^2} \\
l &\approx \frac{g(2.5)^2}{4\pi^2} \\
l &\approx 1.55 \text{ m}
\end{aligned}$$

2.11 Astrophysics and Cosmology

2.11A Gravitational Fields

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Given the following properties of the Earth:

$$m = 5.9 \times 10^{24} \text{ kg} \quad (2.7)$$

$$r = 6.371 \times 10^6 \text{ m} \quad (2.8)$$

Exam Practice 11A p. 177, Question 3

The Orion spacecraft consists of a capsule (mass = 8913 kg) and a service module (mass = 12 337 kg) and is designed to carry four astronauts (mass 80.0 kg each). Calculate the minimum energy needed to move the fully manned spacecraft completely outside the influence of the gravitational field of the Earth.

- Mass of the Earth = $6.00 \times 10^{24} \text{ kg}$
- Radius of the Earth = $6.40 \times 10^6 \text{ m}$
- $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

A $2.12 \times 10^3 \text{ J}$

B $6.30 \times 10^6 \text{ J}$

C $1.34 \times 10^{12} \text{ J}$

D $1.36 \times 10^{12} \text{ J}$

Total mass $m = 8913 + 12337 + 4 \times 80 = 21\,570$ kg.

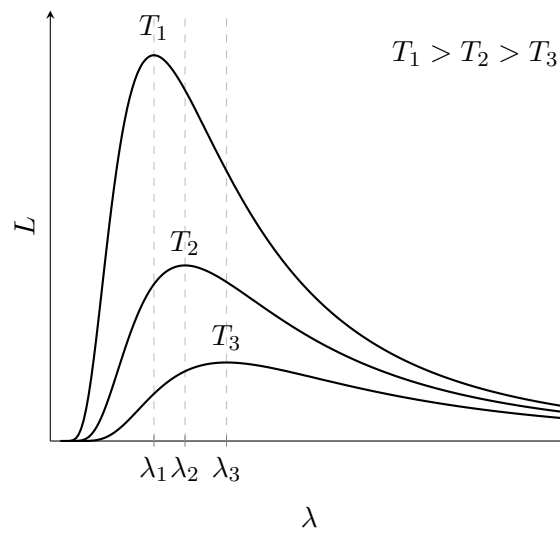
$$\begin{aligned} E_p &= \frac{GMm}{r} \\ &= \frac{(6.67 \times 10^{-11})(6.00 \times 10^{24})(21570)}{6.40 \times 10^6} \\ &= 1.35 \times 10^{12} \text{ J} \end{aligned}$$

Hence the answer is C (closest).

2.11B Space

Wien's Law

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$$\lambda_1 T_1 = \lambda_2 T_2 = \lambda_3 T_3 = \lambda_{\max} T = 2.898 \times 10^{-3} \text{ m K} \quad (2.9)$$

Chapter 3

Other working out

Handout: Gravitational Fields – Questions by Topic, Question 1

Olympus Mons is the highest mountain on Mars. The height of Olympus Mons is 22 km, which is 0.6% of the radius of Mars.

The change in gravitational field strength from the bottom to the top of Olympus Mons is

A -0.6%

B -1.2%

C +1.2%

D +0.6%

Change in gravitational field strength:

$$\begin{aligned}\Delta g &= g_2 - g_1 \\ &= \frac{GM}{r_2^2} - \frac{GM}{r_1^2} \\ &= GM \left(\frac{1}{r_2^2} - \frac{1}{r_1^2} \right) \\ \frac{\Delta g}{g_1} &= \Delta g \cdot \frac{r_1^2}{GM} \\ &= r_1^2 \left(\frac{1}{r_2^2} - \frac{1}{r_1^2} \right) \\ &= \frac{r_1^2}{r_2^2} - 1 \\ &= \frac{r_1^2 - r_2^2}{r_2^2}\end{aligned}$$

Calculate the radii:

$$\begin{aligned}
r_1 \times 0.006 &= 22 \times 10^3 \\
r_1 &= \frac{22 \times 10^3}{0.006} \\
r_1 &= 3.67 \times 10^6 \text{ m} \\
r_1 + 22 \times 10^3 &= r_2 = 3.69 \times 10^6 \text{ m}
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{\Delta g}{g_1} &= \frac{r_1^2 - r_2^2}{r_2^2} \\
&= \frac{(3.67 \times 10^6)^2 - (3.69 \times 10^6)^2}{(3.69 \times 10^6)^2} \\
&= -0.0109
\end{aligned}$$

Appendix A

List of Equations

A.0A Waves

- $v = \lambda f$ Force of tension
- $v^2 = \frac{T}{\mu}$ in a uniform string/rope
Linear density
in kg m^{-1}

Interference:

- p.d. = $n\lambda$ ← constructive
- p.d. = $(n + \frac{1}{2})\lambda$ ← destructive

A.0B Photoelectric Effect

- $h = 6.63 \times 10^{-34} \text{ J s}$
- $E = hf$
- $hf = \phi + E_{k\text{max}}$
- $V = \frac{W}{Q} \iff W = QV$

Appendix B

List of Definitions

B.0A Circuits

e.m.f.

- Energy supplied per unit charge
- Work done per unit charge
- The work done moving unit charge around the whole circuit

Appendix C

Physics Year 2 Textbook

The textbook pages are omitted in draft mode.